

Programming Assignment 3: Denoising Diffusion Probabilistic Models (DDPM)

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1 Introduction

This assignment involves implementing a Denoising Diffusion Probabilistic Model (DDPM) and its deterministic counterpart, DDIM. The Unet architecture and the cosine noise schedule were provided in the assignment starter code. The goal was to train the model on the MNIST dataset for 30 epochs and evaluate its performance through MSE loss plots and generated samples.

2 Implementation Details

The missing parts in `train.py` and `DDPM.py` were implemented, including:

- The forward diffusion process using a cosine schedule.
- The training loop logging MSE loss.
- Sampling procedures for both DDPM and DDIM with various steps.
- Model evaluation using qualitative (samples) and quantitative (MSE) measures.

3 Training Results

The DDPM model was trained on the MNIST dataset for **30 epochs**. The Mean Squared Error (MSE) loss was logged during training. The following plot illustrates the loss trend over epochs.

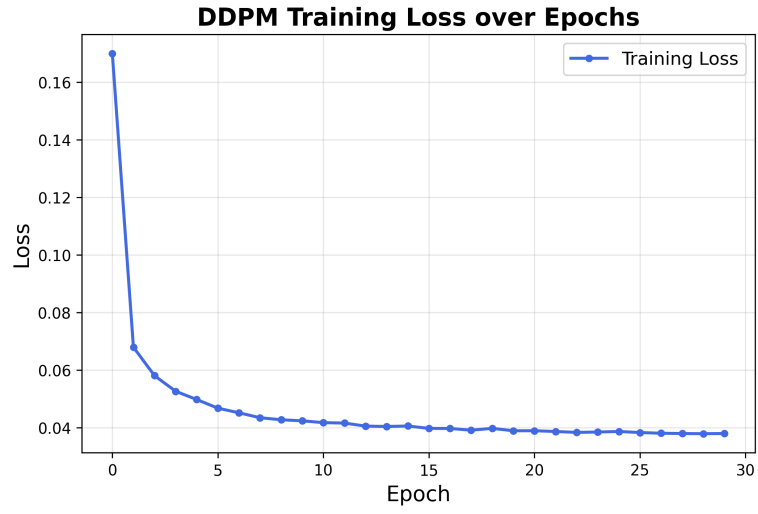


Figure 1: Training loss curve over 30 epochs. The decreasing trend indicates proper learning.

4 Generated Samples from DDPM

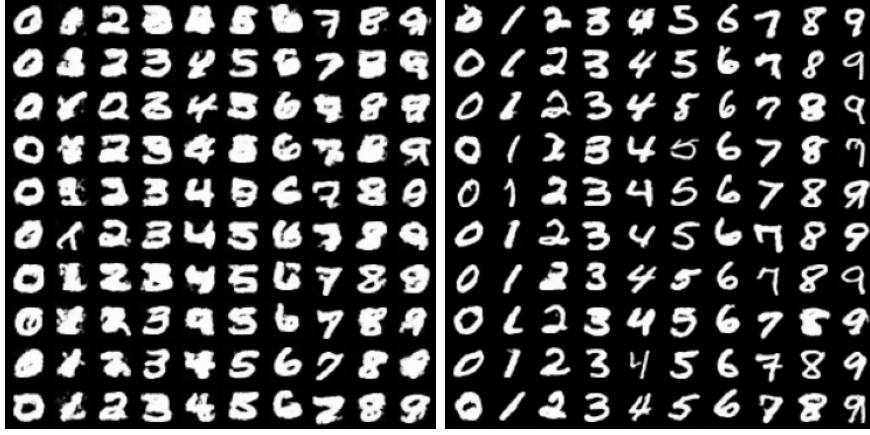
After training, samples were generated using the DDPM model with **1000 timesteps**. The generated images are shown below.



Figure 2: Generated samples from DDPM after 30 epochs with 1000 timesteps.

5 DDIM Sampling with Different Steps

DDIM sampling was implemented using the trained DDPM model. The following figure presents samples generated with different numbers of steps (5, 10, 20, and 50).



(a) DDIM with 5 steps

(b) DDIM with 10 steps



(c) DDIM with 20 steps

(d) DDIM with 50 steps

Figure 3: Generated samples using DDIM with different timesteps (5, 10, 20, 50). Lower steps show lower-quality images.

6 Conclusion

The DDPM model successfully learned to generate MNIST-like digits over 30 epochs. The MSE loss plot showed consistent learning, and the generated samples from DDPM and DDIM demonstrated the effectiveness of the denoising process. DDIM with lower steps produced less refined samples, confirming the importance of diffusion steps in quality generation.